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METHODS AND APPARATUS FOR MINIMALLY INVASIVE BUNION SURGERY

Abstract

A system and method for performing a percutaneous bunionectomy. The system including a jig, which subsequent to attachment and performance of an osteotomy on the bone, uses a rack and pinion mechanism to translate the separated portion of the bone to a surgeon determined displacement λ . A locking mechanism maintains the desired displacement of the bone portions and allows the surgeon, with use of an attached visual guide, to insert a k-wire through the bone portions to serve as a guide wire for the internal fixation that maintains the displacement during the healing. The system removes the requirement that the surgeon manually maintain the desired displacement while inserting the guide wire.

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Background/Summary

CROSS-REFERENCE TO THE RELATED APPLICATION [0001] This application is a continuation filed under 37 C.F.R. § 1.53 claiming the benefit under 35 U.S.C. § 120 of any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application, including U.S. patent application Ser. No. 17/933,640, filed Sep. 20, 2022, and U.S. Provisional Patent Application No. 63/264,246, filed Nov. 18, 2021, and are hereby incorporated by reference in accordance with 37 C.F.R. §§ 1.57; 1.97; and 1.98 in their entireties.

TECHNICAL FIELD

[0002] The disclosed subject matter relates generally a guide system and method to aid an orthopedic surgeon in translating the distal bone fragment and inserting a k-wire into the bone.

BACKGROUND

[0003] Minimally invasive bunion surgery, also known as percutaneous bunionectomy, is a surgical procedure that seeks to correct a bunion deformity by utilizing a small incision. Hallux Valgus is a deformity of the big toe joint wherein a deviation of the big toe towards the lesser toes results in a prominent bump, or bunion.

[0004] During a minimally invasive bunion procedure, a surgeon will create an osteotomy in the first metatarsal and then translate a portion of the metatarsal longitudinally prior to installing fixation e.g., Kirschner wire (“k-wires”). Prior art methods require one instrument to be used for translation of the distal bone fragment and another to be used for k-wire targeting. Maintaining adequate translation while applying and using the targeting guide can be cumbersome and represents a challenge to the surgeon.

[0005] During any surgery, and minimally invasive surgery in particular, surgeons utilize K-wires to both stabilize the position of bones and to serve as guide wires for screw insertion. It is not uncommon for surgeons to require multiple k-wire insertions before they place the guide wire into the desired position. However, repeated k-wire insertions require time, increase frustration during the procedure, and represent micro-injuries to the bone that the body must heal in addition to the surgical correction.

[0006] Current guides/jigs for use in MIS bunion surgery require the surgeon to continuously hold the translation instrument in place in order to maintain the desired translation while targeting the fixation wires. One example, US Patent Application Publication 2020/0060698 entitled SCREW TARGETING GUIDE SYSTEM AND METHOD describes a handheld targeting guide for MIS bunion surgery, the entirety of which is herein incorporated by reference. The current subject matter presents a jig that combines both features into one instrument and independently maintains the desired translation while the surgeon places the fixation wires.

[0007] Reducing the number of instruments, maintaining the desired translation, and enabling the surgeon's hands to focus on k-wire insertion would be beneficial to both the surgeon and patient. Thus, given the deficiencies in the prior art, there remains a need for an improved apparatus to aid in MIS bunion surgery.

SUMMARY

[0008] The embodiments described herein are directed to a system, apparatus, and method for bunion correction utilizing a guide for translating the metatarsal head with respect to the metatarsal shaft subsequent an osteotomy and guiding a k-wire utilized in the fixation of the head and shaft in

the relative translated position. In addition to or instead of the advantages presented herein, persons of ordinary skill in the art would recognize and appreciate other advantages as well.

[0009] In some embodiments, a system for performing a percutaneous bunionectomy is provided. The system including a jig having a body, a shaft contact pad connected to the jig body and conformed to the soft tissue proximate the metatarsal shaft, an arm extending in a longitudinal direction from the jig body, a metatarsal head contact pad connected to the arm and configured to conform to the soft tissue proximate the metatarsal head; a track rigidly connected to the second arm end and movably connected to the jig body and a circular track cooperating with an engagement mechanism of the track, such that a rotation of the track drive results in a lateral movement of the track. The system further includes an alignment guide arm connected to and extending longitudinally from the jig body and a plurality of positioning k-wires, attaching each of the metatarsal head contact pad and the metatarsal shaft contact pad through a multi-dimensional fixation hole to the metatarsal bone. A guide wire is also provided in the system for insertion through the displaced shaft and head of the metatarsal to assist in the subsequent insertion of a fixation screw.

[0010] In some embodiments, a method is provided, the method including the steps of creating an incision in the soft tissue of the foot, positioning a jig upon the soft tissue surrounding the metatarsal bone; securing the jig to the metatarsal bone and, via the incision cutting the metatarsal bone, creating a first portion of the metatarsal and a second portion of the metatarsal. The method further includes translating the jig track to laterally displace the second portion with respect to the first portion; adjusting the visual alignment guide to a desired trajectory; and inserting a fixation k-wire as a screw guide through the first and second portions of the metatarsal such that a fixation screw may permanently fix the bone portions in the laterally displaced position. In the method, the jig is subsequently removed from the metatarsal and the incision is closed.

[0011] In yet another embodiment, a translation and insertion jig for insertion of a k-wire is presented. The jig having a body, a shaft contact pad connected to the jig body and conformed to the soft tissue proximate metatarsal shaft, an arm extending in a longitudinal direction from the jig body, a metatarsal head contact pad connected to the arm and configured to conform to the soft tissue proximate the metatarsal head; a track rigidly connected to the second arm end and movably connected to the jig body and a track drive cooperating with an engagement mechanism of the track, such that a rotation of the track drive results in a lateral movement of the track. The system further includes a lock configured to selectively restrain the track from translation relative to the jig body and an alignment guide arm connected to and extending longitudinally from the jig body

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The features and advantages of the present disclosures will be more fully disclosed in, or rendered obvious by, the following detailed descriptions of example embodiments. The detailed descriptions of the example embodiments are to be considered together with the accompanying drawings wherein like numbers refer to like parts and further wherein:

[0013] FIG. 1 is an isometric illustration of a translation and insertion jig in accordance with some embodiments of the disclosed subject matter;

[0014] FIG. 2 illustrates the translation and insertion jig positioned on the soft tissue near the metatarsal bone of the foot in accordance with some embodiments of the disclosed subject matter;

[0015] FIG. 3 illustrates the translation and insertion jig secured to the metatarsal bone via k-wires in accordance with some embodiments of the disclosed subject matter;

[0016] FIG. 4 illustrates the translation and insertion jig attached to the metatarsal bone subsequent an osteotomy of the metatarsal bone and prior to the lateral displacement of the metatarsal bone

head via the jig in accordance with some embodiments of the disclosed subject matter;
[0017] FIG. 5 illustrates the translation and insertion jig attached to the metatarsal bone subsequent to the lateral displacement of the metatarsal bone head via the jig in accordance with some embodiments of the disclosed subject matter;
[0018] FIGS. 6A and 6B illustrate in a lateral and A-P view, respectively, the insertion of a guide wire into the metatarsal shaft and laterally displaced head via a visual guide on the translation and insertion jig in accordance with some embodiments of the disclosed subject matter;
[0019] FIG. 7 illustrates the track drive and released lock in accordance with an embodiment of the disclosed subject matter; and
[0020] FIG. 8 is a flow chart of a method of performing MIS bunion surgery with the translation and insertion jig in accordance with embodiments of the disclosed subject matter.

DETAILED DESCRIPTION

[0021] The description of the preferred embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description of these disclosures. While the present disclosure is susceptible to various modifications and alternative forms, specific embodiments are shown by way of example in the drawings and will be described in detail herein. The objectives and advantages of the claimed subject matter will become more apparent from the following detailed description of these exemplary embodiments in connection with the accompanying drawings.

[0022] It should be understood, however, that the present disclosure is not intended to be limited to the particular forms disclosed. Rather, the present disclosure covers all modifications, equivalents, and alternatives that fall within the spirit and scope of these exemplary embodiments. The terms “couple,” “coupled,” “operatively coupled,” “operatively connected,” and the like should be broadly understood to refer to connecting devices or components together either mechanically, or otherwise, such that the connection allows the pertinent devices or components to operate with each other as intended by virtue of that relationship.

[0023] FIG. 1 illustrates a translation and insertion jig **100** for minimally invasive bunion surgery. The jig includes a body **110** which extends from its first end **112** configured to face the bone to a second end **114**. The jig **100** includes a metatarsal shaft contact pad **122** connected to the first end **112** of the jig body **110**. The shaft contact pad **122** is preferably conformed to the portion of soft tissue (e.g., skin, muscle, fat, fibrous tissue, blood vessels, or other supporting tissue of the body) covering the foot proximate to the metatarsal shaft to which it is intended to engage (e.g. concave). The contour of the shaft contact pad **122** may be conformed during surgery to match the soft tissue surrounding the bone, or pre-contoured. The shaft contact pad **122** may also be selectable by the surgeon from a plurality of different sizes and shapes, such that the jig **100** may have a set of interchangeable contact pads, or pads that attach to the first end **112**. The shaft contact pad **122** along with the head contact pad **124** discussed later, include at least one k-wire insertion hole **126**, **128** that allows multi-dimensional fixation (i.e., restraint of relative movement in the vertical, lateral and longitudinal directions) of the pads **122**, **124** to the respective shaft and head of the metatarsal bone (as shown in FIG. 3).

[0024] Within or upon the jig body **110** is a linear track **140** extending in the lateral direction, the track **140** rigidly connected to the second end **134** of the extending arm **130** and movably connected to the jig body **110**. The linear track **140** includes an engagement mechanism, which, shown in FIGS. 1-7, is the rack of a rack and pinion mechanism which includes a toothed surface **142**. The track **140** alternatively may have a threaded surface such as in a lead screw mechanism. The track **140** is retained to the jig body to allow only lateral movement. FIG. 4 illustrates a groove **117** in the jig body **110** that retains the track **140** in one embodiment, of course other known retention methods are also envisioned. The track **140** may be incremented to enable the surgeon to ascertain the amount of lateral translation and may be further indexed to move one increment (click) at a time, via indentations and ball stops (not shown), or other indexing mechanism known

in the art.

[0025] The jig **100** has a longitudinal extending arm **130**, a second end **134** of the arm **130** attached to a track **140**. As with the shaft contact pad **122** and the head contact pad **124** discussed later, the extending arm **130** may be contourable, pre-contoured, or modular such that different sizes and shapes may be selected by the surgeon to be attached directly or indirectly to the track **140**. On the other end **132** of the extending arm **130** is a second contact pad **124** for the engagement of the soft tissue surrounding the metatarsal head proximate the pad **124**. The contour of the head contact pad **124** may also be conformed during surgery to a shape that matches the soft tissue proximate the bone (e.g. concave) or pre-contoured. The head contact pad **124** may also be selectable by the surgeon from a plurality of different sizes and shapes, such that the jig **100** may have a set of interchangeable contact pads, or pads inserts that attach to the first end **132** of the arm **130**. The arm **130** may include a longitudinal section **131**, a lateral section **133** and a vertical section **135** to position the head contact pad **124**. The longitudinal section **131** and the vertical section **135** may be adjustable by the surgeon to adjust the position of the head contact pad **124**, or alternatively may be selectable from a plurality of sizes and shapes in the same manner discussed above.

[0026] A track drive **150** (see also, FIGS. **4**, **7**), including a manual knob **155**, dial, or wheel is rotatably connected to the jig body **110**, at least a portion of the knob or dial wheel extending outside the jig body **110** to be accessible for manual engagement. The track drive cooperating with the engagement mechanism of the track **140**. As shown in FIGS. **1** and **7**, the track drive **150** includes a circular gear with intermeshing teeth **157** that functions as a pinion in a rack and pinion mechanism, the rack and pinion may also engage frictionally with high friction surfaces on the pinion **150** and cooperating track **140**. Alternatively, the track drive may be a worm drive, an internally threaded wheel of a lead screw mechanism, or a ball thread drive, each of which translates rotation of the knob, dial or wheel into linear lateral advancement of the track **140** (i.e., rotation of the track drive moves the track **140** and arm **130** laterally. Each increment of rotation of the circular drive (pinion) or length of track (rack) may be biased into position with known methods such as stop points achieved using teeth or ball plungers so that the track clicks into a plurality of positions laterally spaced from the jig body. Additionally, each increment may be of a predetermined measurement such as a mm or fraction thereof.

[0027] A track lock **160** is attached to the jig body **110** configured to selectively restrain the track from translation relative to the jig body **110**. The track lock **160** when engaged prevents the track **140** from retracting laterally and in some embodiments prevents any lateral translation irrespective of direction. The track lock **160** may take the form of a ratchet, allowing the track to advance but not retract, an interlock **162** (as shown in FIG. **7**) in which the lock **160** does not allow any translation. The track lock **160** is preferably biased in a locking position, freeing the surgeon from holding the jig to maintain the bone displacement. The track lock **160** may directly engage the track **140** to prevent its translation, or may engage the track drive **150** preventing it from rotating and thus restrict translating. The track lock **160** may alternatively be a threaded restraint screw (not shown) tightened to frictionally engage either the track **140** or drive **150**, and loosened to release the same, or vice versa. The track lock **160**, shown in FIGS. **3** and **6**, is biased in the locked position by a spring **163** and released by pressing button **164** removing the interlock **162** from the pinion gears **157**, as shown in FIG. **7**. With respect to alternative embodiments in which the track drive **150** is a lead screw or worm drive, the mechanical configuration largely limits the need for a track lock **160** since the track **140** cannot drive the track drive **150**, thus locking the track **140** in place.

[0028] An alignment/targeting guide arm **170** is removably attached to the jig body **110** and extends longitudinally from the jig body **110**. A post **172** extends vertically from an end of the alignment guide arm, and visual alignment rod **174** is attached to the post which is selectively rotatable about a vertical axis with respect to the alignment guide arm **170** for establishing the desired trajectory in the A-P plane. Although not shown a second visual alignment rod which is

selectively rotatable about a horizontal axis with respect to the alignment guide arm may also be beneficial in establishing the desired trajectory in the lateral plane. The post **172** may feature one or more holes **176a,b** aligned with the alignment rod **174**. These holes **176a,b** may be beneficially sized to accept 3.0 mm and/or 4.0 mm MICA Screw k-wires, for insertion into the metatarsal portions using the jig **100**. The visual alignment rod **174** may be created via hole in the vertical post **172** through which a k-wire is inserted to serve as the visual alignment rod **174** when fluoroscopy is used. The alignment guide **170** may also incorporate sleeves through which k-wires are inserted to ensure the wires enter the bone at the proper trajectory.

[0029] FIG. **2** illustrates the jig **100** positioned on the soft tissue (not shown) proximate the metatarsal **210** of a foot **201**. The shaft contact pad **122** is positioned upon the soft tissue covering the metatarsal shaft **211** and the head contact pad **124** is positioned upon the soft tissue covering the metatarsal head **212**. As shown in FIG. **1**, the track **140** and arm **130** are in the retracted or initial position. The extending arm **130** may be configured with respect to the anticipated incision location so as not to interfere with the osteotomy instruments access to the metatarsal bone **210** via the incision. As shown in FIG. **2**, the shape of the extending arm **130** leaves an access area **280** to the metatarsal **210** between 270 and 360 degrees.

[0030] FIG. **3** illustrates the attachment of the jig **100** to the metatarsal bone **210** via k-wires **324** and **322**. The respective k-wires **322** and **324** are inserted through holes **126** and **128** in contact pads **124** and **122** respectively through the soft tissue and into the shaft **211** and head **212** of the metatarsal **210**. The holes **126** and **128** are preferably oriented to direct the k-wires into the bone to ensure multi-dimensional fixation of each of the contact pads **122** and **124** to the shaft **211** and head **212** of the metatarsal bone **210**. The multi-dimensional fixation of each portion of the metatarsal bone **210** is desirable to ensure the jig **100** remains secured to the respective bone portions subsequent the osteotomy.

[0031] In FIG. **4**, the jig **100** positioned on the metatarsal bone **210** is shown subsequent to the osteotomy **410**. As a result of the osteotomy **410**, the metatarsal **210** is divided into a shaft portion **412**, including the shaft **211**, being attached to the shaft contact pad **122** and a head portion **414** including the head **212** being attached to the head contact pad **124**. The track **140** and extending arm **130** are shown in a retracted position or initial position prior to the translation of the head portion **414** with respect to the shaft portion **412** via the jig **100**.

[0032] FIG. **5** illustrates the jig **100** with the track **140**, arm **130** and the head contact pad **124** extended to the desired location selected by the surgeon. As can be seen in FIG. **5**, at the osteotomy **410**, the head portion **414** has been translated laterally from the shaft portion **412** by λ . The track **140** is configured to laterally translate from a first position as shown in FIGS. **2**, **3** and **4** to a fully translated position including any intermediate surgeon selectable position as shown in FIGS. **5** and **6B**. The track **140** is restrained to a predetermined path, typically a straight line, however, it may be beneficial in some circumstances for the path to be curved or not linear.

[0033] Turning to FIGS. **6A** and **6B**, the insertion of the guide wire **601** through the shaft portion **412** and head portion **414** of the metatarsal **210** is shown. With the translation λ established and maintained via the jig **100**, the k-wire **601** is placed through the k-wire hole (**176a,b**) in the post **172** and inserted into the metatarsal **210** following the trajectory established by visual guide rod **174**. A screw may then be guided by the guide wire **601** into the shaft portion **412** and head portion **414** of the metatarsal to fix the translation λ throughout the healing period. Because of the lock **160** or the inherent nature of the track drive **150**, the displacement λ is maintained by the jig **100** without the necessity of surgeon's attention or hands, which are freed for the insertion of the guide wire **601** and ultimately the fixation screw (not shown).

[0034] The visual guide **174** enables the surgeon to visualize the relative position of the surgical k-wire **601** within both portions of the bone **210**. FIG. **6B** illustrates an anterior posterior (A-P) view, i.e. looking down upon the foot **201**. The visual guide rod **174** is coincident with the surgical k-wire **601** when the k-wire **601** is following the prescribed trajectory in the A-P view as shown in FIG.

6B. If a second visual guide is provided, it would be coincident with the surgical k-wire **601** when the k-wire **601** is following the prescribed trajectory in the lateral view, generally shown in FIG. 6A. Thus the location and movement of the inserted k-wire **601** in the A-P view may be represented by the visual guide rod **174**, additionally, the visual guide rod **174** may also be rotatable with respect to the horizon and thus act as an insertion guide in the lateral plane (i.e. the plane normal to the lateral direction), though it requires the surgeon to account for the vertical offset in the lateral view. Likewise, in FIG. 6A, the second guide rod (not shown) may govern movement of the inserted k-wire **601** in the lateral plane. Thus between the top and side views of FIGS. 6A and 6B respectively, the location and trajectory of the inserted k-wire **601** may be represented in three dimensions and thus monitored and controlled using one or two visual guide rods.

[0035] FIG. 8 illustrates a method **800** for performing a percutaneous bunionectomy using the jig **100**. First, as is common in a conventional percutaneous bunionectomy, an incision in the soft tissue of the foot **201** of a patient is created by the surgeon as shown in Block **802**. The jig **100** is positioned upon the soft tissue (not shown) proximate the metatarsal bone **210** of the foot **201** as shown in Block **804** (see also FIG. 2 which illustrates the jig **100** on the soft tissue proximate the metatarsal **210**). As properly positioned, the shaft contact pad **122** and a head contact pad **124** of the jig **100** are proximate the shaft **211** and head **212** respectively of the metatarsal bone **210**.

[0036] In Block **806**, the jig **100** is secured to the metatarsal bone **210** as illustrated in FIG. 3. In FIG. 3, the jig **100** is secured to the bone via k-wires **322** and **324** inserted through the contact pads, soft tissue and into the bone, however it is envisioned other fastening devices (i.e. screws, pins, clamps etc.) may also have beneficial applications.

[0037] In Block **808**, the metatarsal bone is accessed through the incision and is cut creating first and second portions **412** and **414** of the metatarsal **210**, the result of the osteotomy **410** is illustrated in FIG. 4. The jig track **140** is then translated by manually rotating knob **155** of the track drive **150** such that the desired displacement λ is achieved between the two portions as shown in Block **810**. FIG. 5 illustrate the track **140**, arm **130**, head contact pad **124** and head portion **414** of the metatarsal with the desired displacement λ . As noted above, the locking mechanism **160** or the mechanical configuration of the track drive **150** prevents the track **140** from retracting, thus in some embodiments, the locking mechanism must be activated subsequent translation, or in the case where the locking mechanism **160** is biased in a locking position, the locking mechanism would be released to allow the translation of the track **140**.

[0038] As shown in Block **812**, the visual guide rod **174** of the alignment guide **170** is adjusted to the desired trajectory. FIG. 6B illustrates the visual guide rod **174** aligned along the surgeon selected trajectory. In Block **814**, a guide wire is inserted through the first and second portions **412** and **414** of the metatarsal **210** to fix the first and second portions in the laterally displaced position. FIG. 6A illustrates the k-wire **601** inserted through the shaft portion **412** and the head portion **414** when they are displaced by λ . The surgical k-wire **601** is positioned through one of the guide holes **176a**, **176b** and into the soft tissue of the foot **201**. The k-wire **601** is advanced while aligning it with the visual guide rod **174** in the A-P plane, and if a second guide rod is provided, then aligning the k-wire **601** with the other guide rod in the lateral plane. The surgeon may check the position in both A-P and lateral planes to determine trajectory of surgical k-wire and adjust as needed. These latter steps typically require use of fluoroscopy to confirm intended trajectory has been achieved. The surgical k-wire **601** is advanced into the bone while maintaining the desired trajectory in the lateral and A-P planes with reference to the respective visual guides. The surgical k-wire may be advanced incrementally and its position and orientation again verified via the visual guide as it is advanced. As noted, a fluoroscope is preferably used to visualize the bone and the surgical k-wire with reference to the visual guide. Thus, it is preferable that the visual guide rod **174** is not X-ray permeable, so it will appear on the fluoroscope. As shown in Block **816**, the jig **100** is removed from the metatarsal **210**. Subsequent removal of the jig **100**, the surgeon may use a depth gage placed over the guide k-wire **601** to measure screw length. The surgeon then places a drill over the

k-wire **601** with a cannulated drill, then inserts a fixation screw over the k-wire to properly fixate the osteotomy. The guide wire **601** is then removed. A second screw may also be placed lateral to this screw depending on patient anatomy and needs. The incision is closed in Block **818**.

[0039] The visual guide **170**, as discussed above, may have one or more guide rods that extend parallel to the surgical k-wire and serve as representations of the surgical k-wire within the respective planes, to assist in its proper insertion.

[0040] Another aspect of the disclosed subject matter is that the visual guides may be any narrow elongated straight rod, strip or bar that achieve the desired goal of visually referencing the position of the surgical k-wire being inserted. Examples may be rods formed from plastic, ceramic, metal or organic material, or composites thereof. For materials that are x-ray permeable, non-permeable material may be added to enable the guide to be visible under a fluoroscope.

[0041] While the jig **100** shown in FIGS. **1-7**, is configured for the left foot, it is equally envisioned that the jig could be likewise configured for the right foot, or even configured for universal use on both the left and right feet with minor adjustments as understood by one skilled in the art.

Additionally, although the disclosed subject matter is described with respect to a rack and pinion system (i.e. the track is the rack and the drive is the pinion) other configurations are also envisioned, such as a worm drive for translating the track, ball-screw drive system, or a lead screw system. Additionally, while the lock is described with respect to an interlock key, other configurations, such as a frictional lock or ratcheting lock are also envisioned.

[0042] The method shown in FIG. **8**, describes the positioning and securing of the jig **100** prior to the creation of the incision, and/or the osteotomy. However, one or both of the incision and the osteotomy may be performed prior to the positioning and/or securing of the jig **100**. In the case where the osteotomy is preformed prior to the securing of the jig **100**, the jig **100** need not account for access **280** to the bone **210**, other than for the guide wire **601** and subsequent fixation screw.

[0043] Another aspect of the disclosed subject matter is that although the jig **100** is described in view of minimally invasive bunion surgery, it may also be useful in other osteotomies in which portions of a bisected bone are displaced relative to one another, as may be appreciated by those skilled in the art.

[0044] Although the methods described above are with reference to the illustrated flowchart, it will be appreciated that many other ways of performing the acts associated with the methods can be used. For example, the order of some operations may be changed, and some of the operations described may be optional.

[0045] The foregoing is provided for purposes of illustrating, explaining, and describing embodiments of these disclosures. Modifications and adaptations to these embodiments will be apparent to those skilled in the art and may be made without departing from the scope or spirit of these disclosures.

Claims

1. A system for performing a percutaneous bunionectomy, comprising: a jig comprising: a body including a first end and second end; a contact pad connected to the first end and configured to conform to a portion of soft tissue proximate to a metatarsal; an arm extending from the body having a first arm end and a second arm end; a head contact pad connected to the first arm end and configured to conform to a portion of the soft tissue proximate to a metatarsal head; a linear track connected to the second arm end and movably connected to the body, the track including an engagement portion; a circular track drive rotatably connected to the body so as to cooperate with the engagement portion such that a rotation of the circular track drive results in a lateral movement of the track wherein the track may be selectively restrained from translation relative to the body; and a guide arm connected to and extending from the body; a plurality of k-wires, wherein each of the head contact pad and the contact pad, include openings configured to receive one of the

plurality of k-wires and at least one fixation k-wire.

2. The system of claim 1, further including a lock connected to the body and configured to selectively restrain the track from lateral translation relative to the body.

3. An adjustable jig for minimally invasive bunion surgery, comprising: a body having a first body end and a second body end; a metatarsal contact pad connected to the first body end and configured to conform to a portion of soft tissue proximate the metatarsal; an arm extending from the body having a first arm end and a second arm end; a metatarsal head contact pad connected to the first arm end and configured to conform to a portion of soft tissue proximate the metatarsal; a linear track securely coupled to the second arm end while being movably connected to the body, the linear track including an engagement section, a circular track drive rotatably connected to the body and cooperating with the engagement section of the track such that a rotation of the circular track drive results in a lateral movement of the track; and a lock connected to the body and configured to selectively restrain the track from translation relative to the body.

4. The adjustable jig of claim 3, including an alignment guide arm connected to and extending longitudinally from the body.

5. The adjustable jig of claim 3, wherein the engagement section includes a plurality of teeth and the track drive comprises a plurality of intermeshing teeth.

6. The adjustable jig of claim 3, wherein the engagement section includes a high friction surface.

7. The adjustable jig of claim 3, wherein the lock is configured to selectively engage at least one of the engagement section of the track and the circular track drive so as to restrain the track.

8. The adjustable jig of claim 3, wherein the lock is biased to restrain the track.

9. The adjustable jig of claim 3, wherein the metatarsal head contact pad and the metatarsal shaft contact pad are concave.

10. The adjustable jig of claim 3, wherein each of the metatarsal head contact pad and the metatarsal shaft contact pad comprise a k-wire multi-dimensional fixation opening.

11. The adjustable jig of claim 10, including a post extending vertically from an end of the alignment guide arm and a visual alignment rod attached to the post and selectively rotatable about a vertical axis with respect to the alignment guide arm.

12. The adjustable jig of claim 11, including a second visual alignment rod selectively rotatable about a horizontal axis with respect to the alignment guide arm.

13. The adjustable jig of claim 3, wherein the circular track drive engages with the engagement section within the body, the circular track drive including a drive knob, a portion of the knob being external to the body.

14. The adjustable jig of claim 3, wherein the linear track is a rack and the circular track drive is a pinion.

15. The adjustable jig of claim 3, wherein the body further restrains the linear track to a predetermined path.

16. The adjustable jig of claim 11, wherein the post defines a hole for receiving a guide wire.

17. A method for performing a percutaneous bunionectomy, comprising: creating an incision in the soft tissue enveloping the foot of a patient; providing a jig including a body having a first body end and a second body end, a metatarsal contact pad connected to the first body end and configured to conform to a portion of soft tissue proximate the metatarsal, an arm extending from the body having a first arm end and a second arm end, a metatarsal head contact pad connected to the first arm end and configured to conform to a portion of soft tissue proximate the metatarsal, a linear track connected to the second arm end and movably connected to the body, the track including an engagement section, a circular track drive rotatably connected to the body and cooperating with the engagement section of the track such that a rotation of the circular track drive results in a lateral movement of the track with a lock connected to the body and configured to selectively restrain the track from translation relative to the body; positioning a jig upon the soft tissue proximate a metatarsal bone of the foot; securing the jig to the metatarsal bone; translating the jig track to

laterally displace a second portion of the metatarsal bone with respect a first portion of the metatarsal bone; adjusting the visual alignment guide to a desired trajectory; inserting a guide wire through the first and second portions of the metatarsal to fix the first and second portions in the laterally displaced position; removing the jig from the metatarsal; and closing the incision.

18. The method of claim 17, further comprising accessing the metatarsal bone via the incision and cutting the metatarsal bone, creating the first portion of the metatarsal and the second portion of the metatarsal.

19. The method of claim 17, further comprising the steps of installing internal fixation hardware using the guide wire and removing the guide wire prior to closing the incision.

20. The method of claim 17, wherein the step of positioning a jig upon the soft tissue proximate the metatarsal bone of the foot comprises positioning a shaft contact pad and a head contact pad of the jig on the soft tissue of the foot proximate the shaft and head respectively of the metatarsal bone.

21. The method of claim 17, wherein the step of securing the jig to the metatarsal bone comprises inserting a k-wire through a multi-dimensional hole in each of the shaft contact pad and head contact pad, through the soft tissue and into the shaft and head respectively of the metatarsal.

22. The method of claim 17, wherein the step of translating the jig track comprises rotating the drive knob and locking the track into position.

23. The method of claim 17, wherein the step of inserting a guide wire through the first and second portions of the metatarsal, comprises aligning the guide wire with the visual alignment guide.

24. The method of claim 23, wherein the step of inserting a guide wire through the first and second portions of the metatarsal further comprises checking the alignment under a fluoroscope.

25. The method of claim 17, further comprising the steps of installing internal fixation hardware using the guide wire and removing the guide wire prior to closing the incision.
